# Yijun (Asteria) Wang

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**Design Engineering | CAD | Prototyping | Mechanism Design | Human-Centred Design**

Imperial College London Design Engineering student with hands-on experience in CAD, prototyping, and factory layout planning, taking original products from user research through mechanism design and electronics to working prototypes. Strong analytical foundation: 85.91% in Engineering Mathematics, 8th in the cohort for the module.

## Education

**Imperial College London** London, UK — Oct 2025–Jun 2029
MEng Design Engineering — (expected)
- Scored 85.91% in Engineering Mathematics, ranking 8th in the cohort for that module.

**Concord College** Shrewsbury, UK — Sep 2023–Jun 2025
A-Levels: A*A*A*A (Mathematics, Chemistry, EPQ, Further Mathematics)

## Experience

**Vossloh Fastening Systems (China) Co., Ltd. — Production Department Intern** — Jul 2026–Aug 2026
Kunshan, China  Factory layout, AutoCAD, workflow optimisation
- Developed a revised equipment-layout proposal for part of the factory, planning equipment relocation and the addition of a pedestrian walkway to improve space utilisation and movement through the area.
- Evaluated workflow, worker access, ergonomics, and fatigue when positioning equipment, balancing efficient use of space with practical production needs.
- Designed a separate tool-storage plan to improve organisation, visibility, access, and retrieval efficiency.

**Blue Education — Design Engineering Tutor** — Jun 2026–Present
Remote  Admissions, interview preparation, supercurricular and project support
- Support Design Engineering applicants with interview preparation and design projects, giving structured feedback on user needs, iteration, reflection, and project communication.

## Selected Projects

**Bedside ASMR Player — Original Hardware & Software Product** — Aug 2026–Present
User research, mechanism design, closed-loop control, prototyping
- Designed a single-purpose bedside player to address a behavioural problem rather than a technical one, defining the control set around use in the dark and half-asleep so that the device plays a personal library and offers no route to anything else.
- Built a working prototype of the primary control, a motorised fader that both displays and sets playback position, achieving closed-loop accuracy of ±0.30 mm over 100 mm of travel and establishing that the remaining error came from mechanism settling rather than the control loop.
- Integrated a Raspberry Pi with 16-bit ADC position feedback, H-bridge motor drive, $I^2C$ sensing and a Python media pipeline; derived an unlabelled component's pinout by measurement before soldering.

**Food with Asu — Cookbook Web App (Personal)** — Sep 2026–Oct 2026
Product design, UX, AI-assisted build
- Designed a cookbook app turning 100+ saved recipe videos into recipes, weekly plans and a shopping list; directed the build with Claude Code on a Cloudflare backend. In weekly use, holding 119 recipes.

**ACE Workshop Induction Redesign — Service Design & Gamified Training** — 2025–2026
User research, gamified prototype, system design
- Mapped the workshop induction end to end through interviews, observation and journey mapping, then built a scenario-based gamified induction game with knowledge checks, evaluation criteria and a rollout pathway.

**Entia Home Test Packaging Optimisation — Packaging & Commercial Constraints** Jan 2026–Mar 2026
Packaging, cost constraints, concept development
- Developed packaging redesign directions for a home blood-testing device, balancing cost, usability, protection and storage; the team's final design was selected by Entia for future production.

**Asu Desk Tidy — Original Desktop Product Design Project** — Sep 2024–Feb 2025
- Turned user needs for desk organisation into a product concept, then used Fusion 360 and iterative 3D-printed prototypes to test dimensions, support angles and assembly tolerances.

## Technical Skills

**CAD & prototyping:** AutoCAD, Fusion 360, Bambu Studio, 3D printing, soldering and electronics assembly
**Engineering & design:** factory layout planning, spatial optimisation, ergonomics, workflow analysis, human-centred design, product development, mechanism design, closed-loop control, requirements analysis
**Software & electronics:** Python, Linux (Raspberry Pi OS), Git, $I^2C$ and PWM hardware interfacing
**Digital tools:** Figma, FigJam, Excel / Google Sheets, Adobe InDesign, Procreate
**Languages:** Mandarin Chinese (native), English (IELTS 8.0; studying and living in the UK)